Drilling Rig Model and Drilling Processes

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Abstract

Petroleum, sometimes referred to as crude oil or just oil, is a naturally occurring liquid mixture consisting primarily of hydrocarbons that is found in geological formations. Within the energy sources, petroleum is particularly important.

The main technique for extracting petroleum from beneath the surface of the earth is oil drilling, which includes all work done on the well from the start to the delivery (to the production team).

This study aims to precisely assess the environmental impact and characterize the key operations carried out during an oil well's drilling. These are high-risk, financially invested procedures where profitability depends on efficiency and safety above all else. The proper sizing and management of these processes are critical to the success of a drilling operation. In this regard, procedure analysis and operational concerns associated with these processes become crucial for the oil and gas industry and for society. Since oil currently serves as the primary energy source, it aids in the development of best practices and advancing drilling technology. The primary tools utilized in a well's drilling process are discussed first for a better knowledge of all operations; drilling operations are only stated appropriately after that. The drilling program, which includes a detailed description of every operation, is delivered at the end of the well completion process.

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List of Abbreviations

E&P	Exploration and Production	**BOP**	Blowout Preventer
FPSO	Floating Production and Storage and Offloading.	**MPD**	Managed Pressure Drilling
CO2	Carbon Dioxide	**H2S**	Hydrogen Sulfide
PA	Pennsylvania	**bbl**	A barrel of crude oil

1 Chapter One

1.1 Introduction

Fossil fuels like oil and gas come from buried plants and animals, transformed by heat and pressure. Over millions of years, they become "kerogen," which then turns into oil and gas. These were forced upwards through porous rocks until they reached non-porous rocks, resulting in the formation of a larger "reservoir".

The main regions where these reservoirs have been discovered are the Gulf of Mexico, North Sea, Siberia, Middle East, North Africa, Indonesia, Campos/Santos Basins in Brazil, Newfoundland, Nova Scotia, Sakhalin in Russia, the east coast of India and the Caspian Sea [1].

The petroleum industry divides itself into three sectors [1]:

- **Upstream:** This sector includes the Exploration and Production (E&P) phase.
- **Midstream:** Once produced, the hydrocarbons are handed over to the midstream sector that processes, stores and transports the products.
- **Downstream:** This deals with the marketing of finishing products supplied from the midstream sector.

In the upstream sector, an oil exploration and production (E&P) project typically comprises five phases:

1) Exploration
2) Drilling
3) Completion
4) Production
5) Decommissioning

Modern technology has transformed oil and gas exploration. Using satellites and seismic techniques, experts map underground rock formations to look for reservoirs.

For underwater exploration, specialized vessels are used to tow sonar navigation equipment that produces sound waves, also known as shock waves (like compressed air guns), that bounce off rocks. Hydrophones (in water) or geophones (on land) are then used to receive echoes of sound (Figure 1.1). On land, similar methods utilize ground vibrations. Analyzing these echoes and additional data, like gravity variati ons, helps geologists pinpoint areas with a high chance of oil or gas [1, 2].

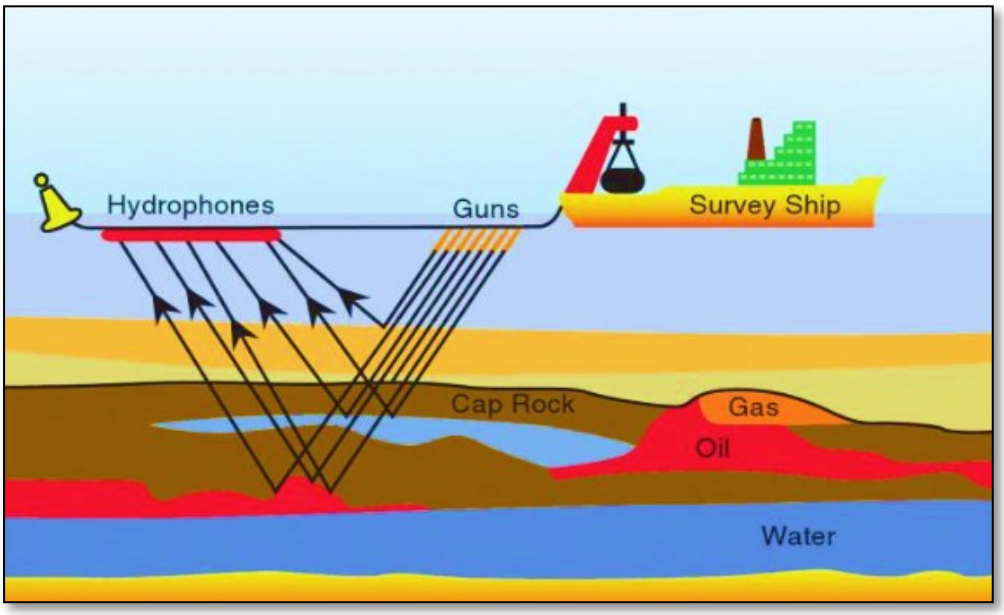

Figure Chapter One .1 Petroleum Exploration

Drilling may be necessary during the exploration phase to determine the precise position and volume of the reservoir. Once developed, the petroleum products must be extracted from the reservoir by larger-scale drilling utilizing "drilling rigs." This study explores the environmental impact and key operations of oil well drilling, both onshore and offshore. It dives into initial exploration drilling to pinpoint oil reservoirs, then analyzes large-scale extraction using rigs, comparing modern and traditional methods. Ultimately, the study aims to comprehensively assess the entire oil well drilling process and its impact.

1.1.1 History of oil well drilling

The history of oil drilling can be traced back to ancient China, where people drilled wells to extract oil and gas as early as the 4th century BC. The first known oil wells were drilled using iron drill bits and bamboo poles. After that, manual pumps or buckets were used to extract the oil. In the 19th century, the oil drilling process began to modernize with the invention of new technologies, such as the cable tool drilling rig and the rotary drilling rig. The cable tool drilling rig used a heavy bit attached to a cable to pound its way into the ground. The rotary drilling rig used a rotating bit to drill through the rock.

The first commercial oil well in the United States was drilled in Titusville, Pennsylvania, in 1859 by Edwin Drake (Figure 1.3). Drake used a combination of cable tools and rotary drilling technology to drill to a depth of 69 feet. His discovery sparked an oil boom in the United States and led to the development of the modern oil industry [3].

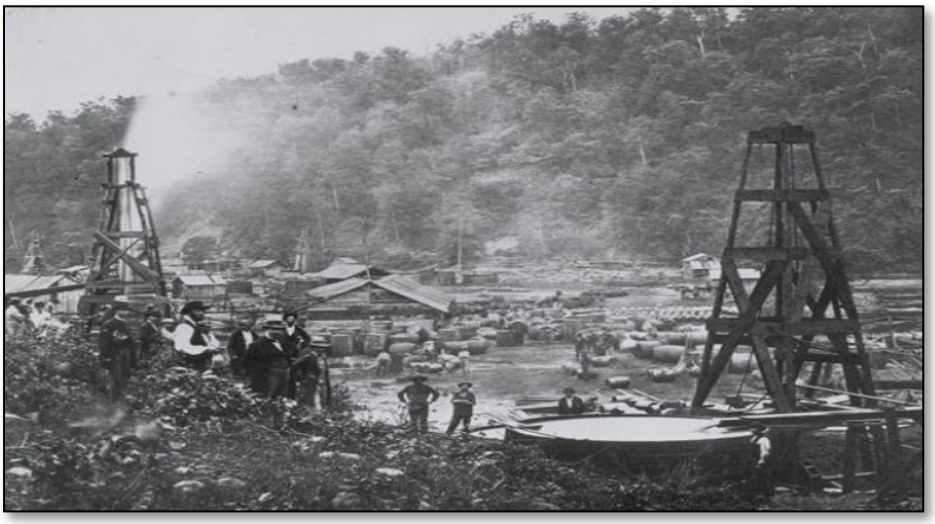

Figure Chapter One .2 Drake Well Museum Collection, Titusville, PA [4].

In the early 20th century, oil drilling technology continued to advance with the development of new tools and techniques. For example, the invention of directional

drilling in the 1920s allowed drillers to drill wells at angles, which made it possible to reach oil deposits that were previously inaccessible.

In recent decades, oil drilling technology has become even more sophisticated. Today, oil companies use a variety of advanced technologies, such as 3D seismic imaging and horizontal drilling, to explore for and produce oil and gas from increasingly challenging environments [3].

The examination of historical safety records and accident data indicates a continued prevalence of environmental incidents. Nonetheless, the impact of these pollutants on both human health and the environment can be significantly attenuated through the implementation of robust environmental monitoring and mitigation strategies. This paper elaborates on the pivotal role of contemporary technological advancements in achieving such outcomes.

1.2 The Aims of the Project

The overall purpose of the research is to present the drilling process, including all its components and advantages. We commence constructing an oil rig in a practical manner. In addition, we are presenting our project to the Runaki institute with the objective of emboldening new students in the department of oil and gas in order to engage in efficient courses and accomplish practicable experience. Besides, we depict drilling techniques and the whole aspect of an oil rig, as well as listing the complete challenges encountered by drilling companies while drilling. Nevertheless, the distinctions between excavation on land and at sea have been considered. In the end, we compared modern and traditional methods of drilling oil wells.

- Focusing on specific rig components (e.g., drilling tools, mud circulation systems) might be more feasible and impactful in a warehouse setting.
- Testing and optimizing innovative drilling tools or technologies developed in a controlled environment.
- Developing novel drilling fluids or mud circulation systems and testing their efficiency and environmental impact within a dedicated facility.

1.3 Report layout (Research organize)

This research composes of five chapters and the layout of this proposition is as following:

Chapter 1: A general introduction to the drilling process and describing the Aim of the project.

Chapter 2: This chapter is dedicated to compiling a wealth of research information concerning the drilling process. Also, the chapter not only addresses the environmental impacts and waste generated during drilling but also explores viable solutions and effective waste management strategies to mitigate and address these concerns.

Chapter 3: This chapter thoroughly explores drilling rig systems and methods, breaking down the key components and operational details. The discussion provides a clear picture of how drilling rigs work and the different approaches used, making it accessible for readers seeking a basic yet comprehensive grasp of the subject.

Chapter 4: This chapter delves into the construction of a drill rig, demystifying its key components and meticulously outlining the assembly process. We embark on a hands-on journey through each stage, equipping you with the technical expertise and practical skills to transform blueprints into a fully functional rig.

Chapter 5: This chapter investigates the intricate relationship between drill rig operations and their environmental consequences. We analyze both established and emerging technologies, scrutinizing their cost-effectiveness and ecological footprints. This critical examination equips you to make informed decisions regarding technology selection, paving the way for a future where drilling progress and environmental responsibility coexist.

2 Chapter Two

2.1 Literature Review

Introduction Oil and gas drilling is a complex process that is used to extract hydrocarbons from the earth's crust. It is a vital sector that has a significant impact on the energy industry, and it contributes significantly to the global economy. The extraction of hydrocarbons has been revolutionized by technology and innovation, and it has resulted in increased productivity and efficiency. This literature review explores recent research on oil and gas drilling processes, presents insights into current practices, and suggests areas for potentially fruitful future research.

2.2 Oil and Gas Disasters and Industrial Hazards Associated with Drilling

In this article [12], the dangers and possibly fatal occurrences associated with onshore and offshore oil and gas drilling operations that have happened over the past eighteen (18) years globally are examined through an exhaustive review of the literature.

The authors draw the conclusion that these operations are extraordinarily dangerous, twice as dangerous as general industries and three times more dangerous than construction, after reviewing a large number of articles and reports.

Two main conclusions may be drawn from the analysis: first, there are disproportionately more accidents and fatalities at drilling sites in Southeast Asia and the Middle East. Second, equipment maintenance emerges as a major source of risk across both onshore and offshore drilling, often due to inadequate safety measures. This demonstrates how tighter laws and better maintenance procedures are required to lessen the risks that drilling crews encounter on a yearly basis.

2.3 Facts and Data

The study of the environmental effects of oil and gas drilling in [13] is done with an eye toward the projects' economic benefits, according to industry experts. Environmental dangers are present both now and in the future with oil and gas exploration, particularly directional and offshore drilling.

The main causes of these concerns are :

(a) contamination from drilling wastes (muds, produced waters, byproducts, etc.);
(b) spills and leaks of natural gas and oil (sometimes blowouts);

(c) direct impacts on both human and wildlife health.

Toxic substances are present in the drilling fluids that are pumped into the well hole, including lead, mercury, oil and grease, arsenic, chromium, cadmium, and naturally occurring radioactive compounds. The location, depth, and kind of drilling fluid all affect the makeup of the produced waters and drilling muds.

Produced waters that may have an effect on groundwater or the surface are normally disposed of in deep aquifers, but the risk remains. Moreover, aquatic organisms are at risk from this.

Potential releases of hydrogen sulfide (H2S) from natural gas deposits also contribute to air pollution. It may have a serious short- and long-term direct impact on human health, ranging from unconsciousness to rapid death. According to statistics, 0.5–1% of exploratory wells result in blowouts, causing harmful emissions.

Previous safety records and accident data suggest that these kinds of environmental incidents will persist. However, the effects of these pollutants on human health and the environment can be minimized or mitigated if proper environmental monitoring and mitigation measures, including the use of modern technological advances

discussed in this paper, are put in place. Reduced air pollution, improved monitoring, recycling, and management of generated wastes to prevent loss or pollution, resource preservation, and on-site recycling of energy byproducts are all results of improved management and advanced technologies for exploration, drilling, production, and oilfield operations. These measures also improve the environment and lower costs by reducing waste and unnecessary pipelines.

2.4 Sedimentation of solids operations

Whereas Santos et al. conducted a study in [14] to address the sedimentation of solids in drilling fluids used in oil well drilling operations. The goal of the study was to comprehend how solid-fluid interactions impact sedimentation and pressure, with a particular emphasis on areas with high solid concentrations.

The addition of calcium carbonate increased the stability of the drilling fluids studied in the laboratory. Compared to non-thickened fluids, fluids containing calcium carbonate showed slower sedimentation, less compressible sediments, and increased pressure on solids.

In general, drilling fluids perform better and are more stable when calcium carbonate is added. This increases the effectiveness and safety of drilling operations for oil wells, particularly in areas with significant concentrations of solids.

This suggests that adjusting the formulation of drilling fluids based on operational conditions is essential for effective wellbore management.

2.5 Introduction to Smart Oil and Gas Wells

The objective of this investigation [15] is to provide the reader with an overview of "Smart Wells" technology, starting with drilling operations and progressing to completion and continuous monitoring systems.

The paper offers a thorough overview of the technology related to the drilling, completion, and monitoring of smart wells. In order to increase the profitability of unconventional reservoirs, "smart well" technologies are required. The development initiatives are driven by smart wells to be more economically viable and to raise their net present value. That can be achieved by reducing the number of drilled wells, workovers, and intervention operations.

By directly connecting remote areas of the formation to the wellbore, fishbone technology facilitates the creation of a bigger drainage area. The ribs aid in increasing exposure, which raises production from a single well and lowers the number of wells. Drilling technologies of that type can be applied to fields to improve their net present value and redevelop them.

Smart technology can postpone the breakthrough of gas and water and manage the production and injection zones to maximize the well's total productivity. Additionally, they can prevent the water and gas coning and sand production, which raises the well's output.

Furthermore, one of the main benefits of smart technologies is their ability to gather data while the well is in operation without requiring a shutdown for intervention or data collection campaigns.

2.6 An overview of management, recycling, and wasting disposal in the drilling operation of oil and gas wells in Iran

In [16], a team of researchers presents their work, "An overview of management, recycling, and wasting disposal in the drilling operation of oil and gas wells in Iran,".

Drilling operations in oil and gas wells produce a significant amount of cutting. Solid particle depletion in the solid control apparatus is accounted for as a significant amount of waste.

Remaining waste particles in the system can have diverse effects on mud function and the occurrence of potentially destructive impacts on environmental and ecological systems.

Solid-state control devices are used in the new systems' functioning to segregate drilling cuttings. After a solid-state refurbishment process and an inspection of each of their properties, they are buried in a permanent and solid state.

Moreover, every test is conducted in an environmental liquid phase, and the results are recycled, reused, or buried properly.

This extensive study's goal is to present the new wastewater management design, layout, equipment, and planning from an environmental perspective. It is based on information currently available about a number of drilled wells in Iran that are utilized for drilling waste management and solid-state control systems.

As a result, the drilling sector will greatly benefit from the application of new technologies in waste management and solid control since they will enable it to expand its workforce in a number of cost-effective settings.

3 Chapter Three

3.1 Theory of the topic

During the exploration stages, some drilling may be required to establish the exact location and size of the reservoir. Once established, larger-scale drilling by using 'drilling rigs' is required to extract the petroleum products from the reservoir. The stage at which an oil well is readied for extraction is known as completion but the actual stage in which the petroleum contents of the reservoir are extracted is known as production [1].

In the petroleum industry, drilling is an essential procedure that is required to extract oil and gas from underground sources. In order to access the target formation holding the hydrocarbons, a borehole—a long, narrow cylindrical opening—must be created that penetrates the Earth's surface [8].

A cylindrical tool called a drill bit is used in the drilling process to create holes in a workpiece as it revolves. The drilling operation process involves three stages: the start and centering stage, the full drilling stage, and the breakthrough stage. In the first stage, the exact position of the hole is required, whereas the second stage leads to the full engagement of the drill bit, while the last stage includes passing the drill through the underside of the workpiece, where the operation stops [9].

By this time, the drilling rigs are replaced by production platforms that extract petroleum products and distribute them to land-based networks through pipelines, or FPSOs (Floating Production, Storage and Offloading) (Figure 3.1). When the contents of the reservoir are reduced, injection wells may be used to inject water, CO2 or steam to push the required contents out. [1]

Finally, the reservoir may become financially unviable when it has dried and, hence may need to be abandoned. At this stage, the production rig may need to be decommissioned.

Figure Chapter Three .3 An early FPSO Castellon, at the Castellon field offshore Spain in 1976 [10].

Two primary methods dominate the drilling landscape: cable tools and rotary. Each method has been around in ancient times, employs distinct principles and excels in different scenarios. Let's delve into their unique approaches [20].

3.1 Types of oil and gas drilling

The search for oil and gas, two commodities powering the modern world, takes us to diverse landscapes. While vast deserts and arctic tundra's host onshore drilling operations (Figure 1.4), the deep blue expanse of oceans becomes the stage for offshore endeavors (Figure 1.5). Both methods, despite their seemingly distinct settings, share the ultimate goal of accessing these valuable resources hidden beneath the Earth's surface. Delving into the intricacies of onshore and offshore drilling unveils a fascinating tale of engineering prowess, environmental considerations, and the constant push for resource security [5].

- **Onshore Drilling**

Onshore drilling, as the name suggests, occurs on land. The process involves erecting a drilling rig – a towering structure resembling a mechanical giraffe – that bore deep into the Earth's crust. This rig houses a drill string, a long, segmented pipe equipped with a bit at its end. The bit, resembling a toothed cookie cutter, pulverizes rock formations, creating a pathway for the drill string to descend further. As the drilling progresses, mud is pumped down the drill string to cool the bit, lubricate the hole, and prevent the well from collapsing. Once the target reservoir containing oil or gas is reached, the well is cased with steel pipes to ensure stability and prevent contamination. Finally, production equipment is installed to extract the hydrocarbons and transport them to processing facilities.

Onshore drilling can be broadly categorized into two types: conventional and unconventional. Conventional drilling targets well-defined geological formations where oil and gas are naturally trapped. Unconventional drilling, on the other hand, focuses on extracting hydrocarbons from shale formations, which require techniques like hydraulic fracturing (fracking) to break open the rock and release the trapped resources [5, 6].

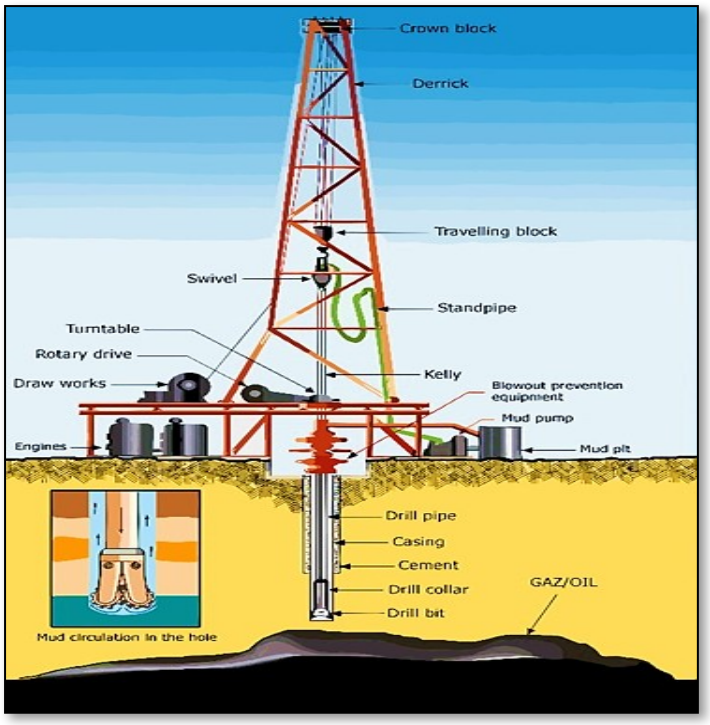

Figure Chapter Three .4 onshore drilling

Offshore Drilling

Here, massive platforms or rigs become temporary islands amidst the waves, serving as the base for drilling operations. These rigs can be fixed structures anchored to the seabed or mobile units capable of operating in various locations. Just like their onshore counterparts, offshore rigs utilize drill strings and mud circulation to create and stabilize wells. However, the marine environment presents unique challenges. The constant motion of waves, the corrosive nature of saltwater, and the immense pressure at deeper depths necessitate robust platform designs and specialized equipment.

Offshore drilling can be further classified into shallow-water and deep-water operations. Shallow-water drilling typically occurs in depths up to 300 meters, where fixed platforms resting on the seabed are commonly used. Deep-water drilling, on the other hand, tackles depths exceeding 1,500 meters, often employing floating

production, storage, and offloading (FPSO) vessels. These vessels extract, process, and store oil or gas before offloading it to tankers for transportation [7].

Figure Chapter Three .5 offshore drillin

- **Comparing Onshore and Offshore Drilling**

While both onshore and offshore drilling share the objective of resource extraction, several key differences set them apart [8, 9]:

a) **Accessibility:** Onshore drilling is generally easier and cheaper to initiate due to readily accessible locations and simpler logistics. Offshore operations, on the other hand, require substantial investments in platform construction, specialized equipment, and skilled personnel.

b) **Environmental Impact:** Onshore drilling can disrupt land use, contaminate soil and water sources, and affect wildlife. Offshore drilling poses risks of oil spills, which can devastate marine ecosystems and coastal communities. Both methods require careful environmental assessment and mitigation strategies.

c) **Production Capacity:** Onshore wells tend to have higher production rates compared to offshore wells, making them more commercially attractive in the

short term. However, offshore fields often hold larger reserves, offering long-term production potential.

d) **Safety Considerations:** Both onshore and offshore operations involve inherent risks for workers, including exposure to hazardous materials, accidents, and extreme weather conditions. Offshore drilling adds the complexity of working in a dynamic marine environment, necessitating additional safety protocols and emergency response measures.

1.3.1 Drilling Problems, Causes, and Solutions

The oil and gas industry are recognized as one of the most hazardous industries on earth. Extracting hydrocarbon from an underground reservoir is very risky and uncertain. Therefore, it is very important to find out the root causes of its risk and uncertainty. The majority of the risks and uncertainties related to this business are encountered while drilling. As a result, drilling problems offer an excellent benchmark for other practices in petroleum engineering as well as other disciplines. However, the key to having a successful achievement of the drilling objectives is to design drilling programs based on anticipation of potential drilling problems. The more comprehensive the list of problems the more accurate the solution manual will become. The best modus operandi is to avoid running into a scenario where problems arise. This preventative style will lead to safer and more cost-effective drilling schemes. It is well understood that even one occurrence of the loss of human life, environmental disaster, or loss of rig side area can have a profound effect on the welfare of the entire petroleum industry. Some of the drilling problems comprise of drill pipe sticking, stuck pipe, drill string failures, wellbore instabilities, hole deviation and well path control, mud contamination, kicks, hazardous and shallow gas release, lost circulation, formation damage, loss of equipment, personnel, and communications. There are some other problems specifically related to slim hole drilling, coiled tubing drilling, extended reach drilling, and under-balance drilling, etc. There is a famous saying, “prevention is better than cure”. So, the motto should

be “drill a hole safely without having any accident, incident, or harm to this planet, with minimum costs”. The drilling operations should be in a sustainable fashion where the minimization of drilling problems and costs has to have the top priority [10].

1.3.2 Application of Drilling Engineering

Throughout human civilization, drilling in numerous forms played a significant role. As such, the applications of drilling technology are numerous. The applications of drilling range from children’s toys to modern drilling of a hole for the purpose of any scientific and technological usage. Humans have been using this technology for underground water withdrawal from ancient times. Drilling technology is a widely used expertise in the applied sciences and engineering such as manufacturing industries, pharmaceutical industries, aerospace, military defense, research laboratories, and any small-scale laboratory to a heavy industry, such as petroleum. Modern cities and urban areas use the drilling technology to get the underground water for drinking and household use. The underground water extraction by boring a hole is also used for agricultural irrigation purposes. Therefore, there is no specific field of application of this technology. It has been used for a widespread field based on its necessity. Therefore, here the drilling engineering application means a shaft-like tool (i.e., drilling rig) with two or more cutting edges (i.e., drill bit) for making holes toward the underground hydrocarbon formation through the earth layers especially by rotation. Hence the major application of drilling engineering is to discover and produce redundant hydrocarbon from a potential oil field [11].

3.1.1 Cable-tool drilling

The first planned oil well was drilled in 1859 by Colonel Drake at Titusville, Pennsylvania, USA. This well was less than 100 ft deep and produced about 50 bbls/day. The cable-tool drilling method was used to drill this first well. The term cable-tool drilling is used to describe the technique in which a chisel is suspended from the end of a wire cable and is made to impact repeatedly on the bottom of the hole, chipping away at the formation (Figure 3.2). When the rock at the bottom of the hole has disintegrated, water is poured down the hole, and a long cylindrical bucket (bailer) is run down the hole to collect the rock chips [12].

In spite of cable-tool drilling's widespread use in the early days, the system has a couple of drawbacks. One is that cable-tool drillers have to stop periodically drilling and pull the bit from the hole. They then have to run a special basket, a bailer, into the hole to retrieve and remove the pieces of rock, or cuttings, the bit makes [21]. Cable-tool drilling was used up until the 1930s to reach depths of 7500 ft [12].

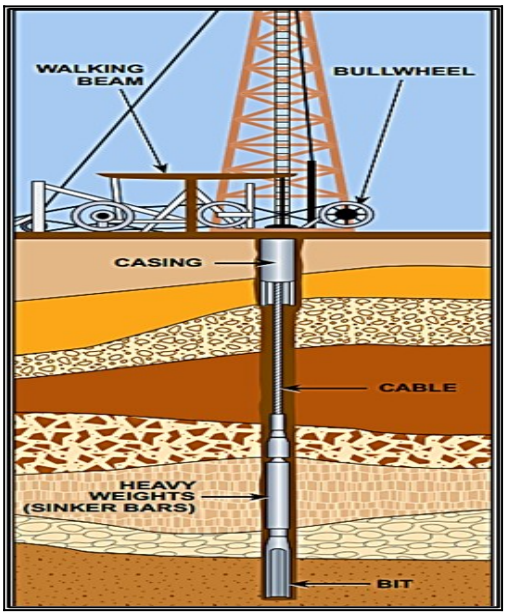

Figure Chapter Three .6 : A cable-tool rig.

3.1.2 Rotary Drilling

In the 1890s, the first rotary drilling rigs (Figure 3.2) were introduced. Essentially, rotary drilling is the process of suspending the rock cutting tool from the end of a hollow pipe so that fluid can flow over the drill bit's face continually, cleaning the drilling material from the bit's face and bringing it to the surface.

This is a much more efficient process than the cable-tool technique. The cutting tool used in this type of drilling is not a chisel but a relatively complex tool (a drill bit), which drills through the rock under the combined effect of axial load and rotation. The first major success for rotary drilling was at Spindletop, Texas, in 1901, where oil was discovered at 1020 ft and produced about 100,000 bbl/day. In 1950, there was an equal number of active cable tool and rotary drilling rigs in the United States. Today, almost all wells are drilled with rotary drilling rigs [12,22].

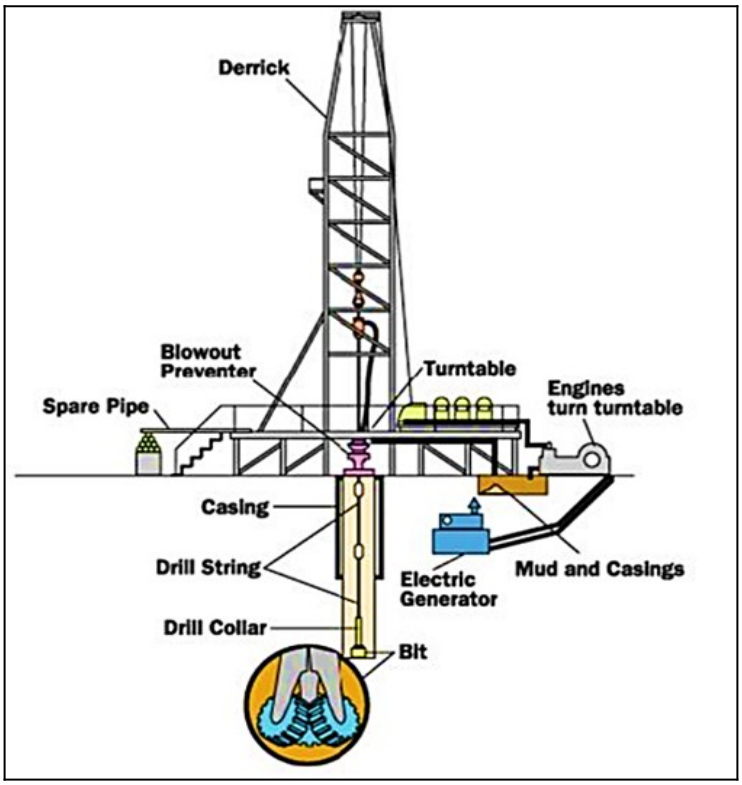

Figure Chapter Three .7 Schematic of a rotary oil rig

3.2 Drilling Rig system

Drilling rigs are the unsung heroes of the energy industry, tirelessly chewing through rock to access valuable resources like oil, gas, and geothermal energy. These complex machines are comprised of several interconnected systems, each playing a crucial role in the drilling process. Although the pieces of equipment associated with these systems will vary in design, they will be found on all drilling rigs. The equipment discussed below will be found on both land-based and offshore drilling rigs [23].

In this chapter, we delve into the five vital systems which are [24]:

1. Power system
2. Hoisting system
3. Rotary system
4. Circulation system
5. Well control equipment

3.3 Power System

The power system is the heart of the drilling rig, supplying the raw energy needed to drive all its operations. In the early days of drilling, steam engines ruled the roost, but diesel engines have since become the industry standard. These powerful engines, often arranged in multiples, burn diesel fuel to generate mechanical power, which is then converted to electricity or used directly to drive various rig components [13, 25]

3.3.1 Components of a Power System

The 'components of power system in drilling rig' include [13, 26]:

- Diesel Engines: The primary source of power, these robust engines can handle the continuous heavy-duty demands of drilling operations.
- Generators: Convert mechanical power from the engines into electricity for powering electric motors and other electrical equipment.
- Driveshafts and Gearboxes: Transmit and adjust power levels to different rig components as needed.
- Control Systems: Monitor and regulate engine performance, ensuring optimal operation and safety.

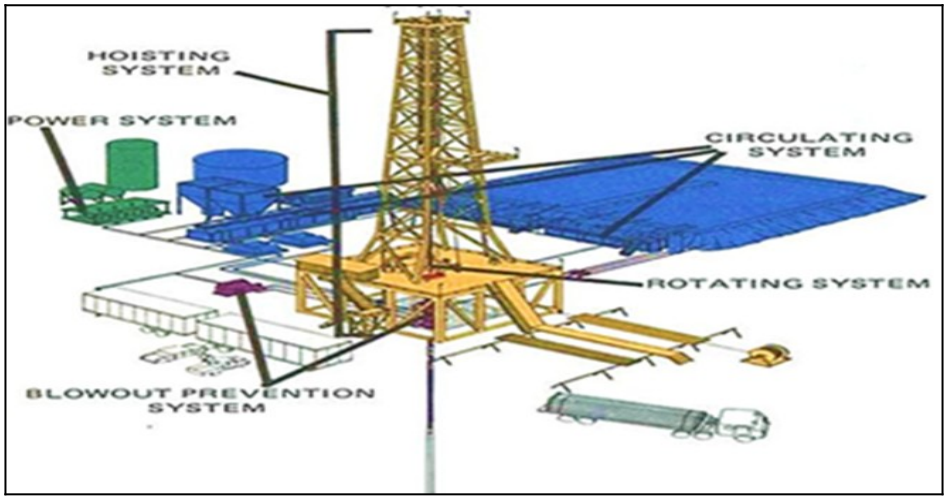

Figure Chapter Three .8 Drilling Rig Power System Layout

3.3.2 Types of Power Systems

Drilling rig power systems are classified into three groups based on their specific needs and location [24]:

- Diesel-Electric: The most common setup, diesel engines generate electricity that powers electric motors throughout the rig. This offers flexibility and ease of control.
- Diesel-Mechanical: Involves direct mechanical transmission of power from the engines to specific rig components like the draw works and mud pumps. This is a simpler system but offers less flexibility.
- Hybrid Systems: Combining diesel engines with electric motors and energy storage options like batteries is gaining traction due to its potential for increased efficiency and reduced emissions.

3.4 Hoisting System

The hoisting system is the muscle of the drilling rig, responsible for raising and lowering heavy equipment through the wellbore.

3.4.1 Types of Hoisting Systems

The most common types of hoisting systems in drilling rigs can be categorized based on their function and primary components [25]:

- Standard Single Drum: The most common type, with one drum for the main hoisting line and another for auxiliary lines.
- Double Drum: Offers greater flexibility and control, with separate drums for the main and auxiliary lines.
- Twin-Drive: Utilizes two drums driven by independent engines, enhancing safety and redundancy.

3.4.2 Components of Hoisting System

The main components of the hoisting system are [27]:

- Drawworks: The workhorse of the hoisting system, the drawworks is a powerful winch that uses steel cables to raise and lower the drill string, casing, and other heavy equipment.
- Crown Block: Mounted at the top of the derrick, the crown block is a system of pulleys that guides and distributes the weight of the load carried by the cables.
- Traveling Block: Attached to the derrick by the cables, the traveling block moves up and down, carrying the drill string and other equipment into and out of the wellbore [13, 23].
- Drill Line/Casing Line: High-strength steel cables for raising and lowering equipment.
- Derrick: Tall structure supporting the crown block and providing clearance for equipment movement.

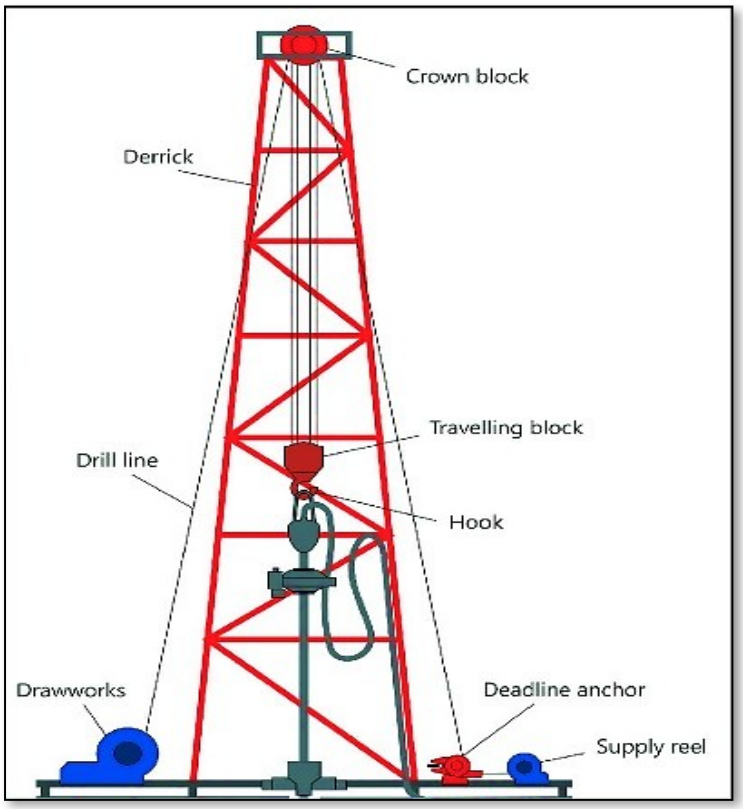

Figure Chapter Three .9 Drilling Rig Hoisting System.

3.5 Rotary System

The rotary system is used to rotate drilling strings in the wellbore and drive the bit to crush the rock. The rotary system of normal drilling rigs consists of a rotary table, Kelly, and drill string [27, 28].

3.5.1 Components of a Rotary System

The rotary system's primary components are as follows:

- Rotary Table: Mounted on the rig floor, the rotary table is a powerful turntable that rotates the Kelly and upper drill string. It's equipped with a bearing assembly designed to handle the immense weight and rotational forces involved [23].
- Drill Bit: Specialized cutting tool mounted on the end of the drill string, designed for specific rock formations [1].
- Drill String: A long, continuous column of heavy-duty steel pipes connecting the kelly to the drill bit. It transmits the rotational force, carries the drilling mud, and provides structural support for the weight of the bit and drilling fluid [23].
- Kelly: A thick-walled, square or hexagonal steel pipe, the kelly transmits the rotary table's rotational force to the drill string. It's connected to the rotary table through a kelly bushing and can be added or removed in sections as the wellbore deepens [1].

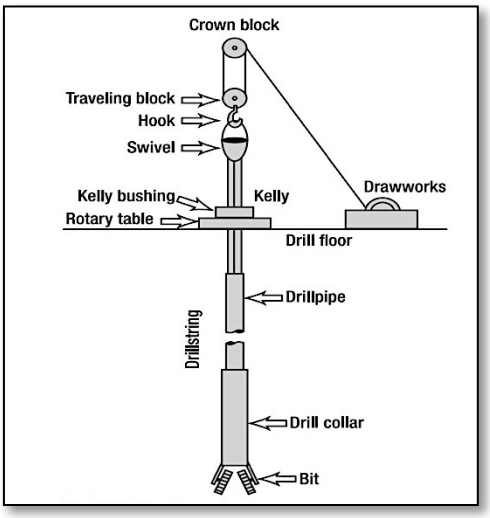

Figure Chapter Three .10 Rotating System

3.5.2 Types of Rotary Systems

There are two primary types of rotary systems used by drilling rigs [28, 29]:

- **Top Drive**: The rotating table is located at the top of the derrick, directly driving the drill string through a rotating kelly joint. This eliminates the need for the kelly and offers greater control and efficiency.

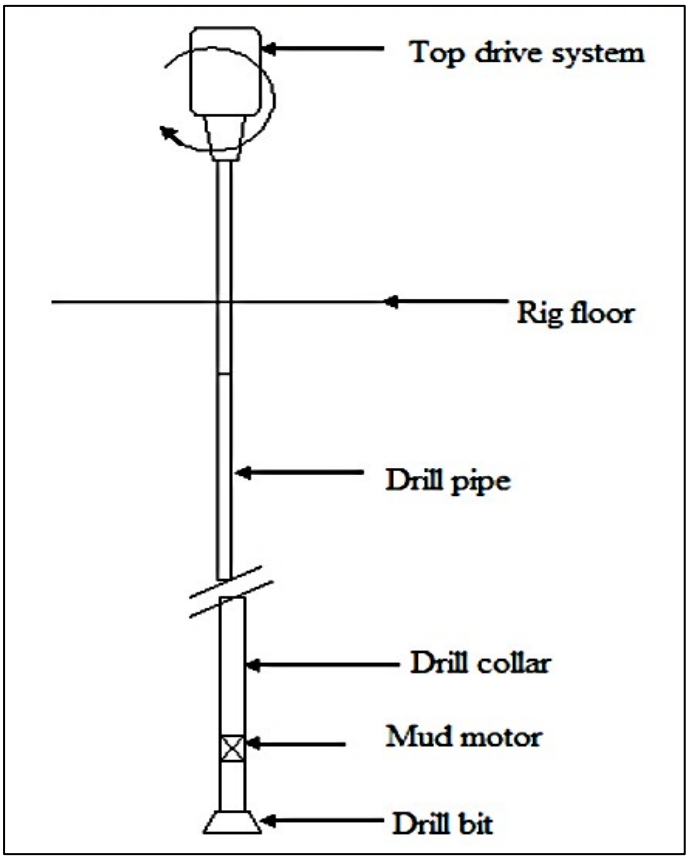

Figure Chapter Three .11 Top drive rig drill string sketch

Conventional Rotary: The traditional setup, with the rotary table at the bottom of the derrick and the kelly connecting it to the drill string. This system is simpler but requires handling and adding/removing kelly sections as the wellbore deepens.

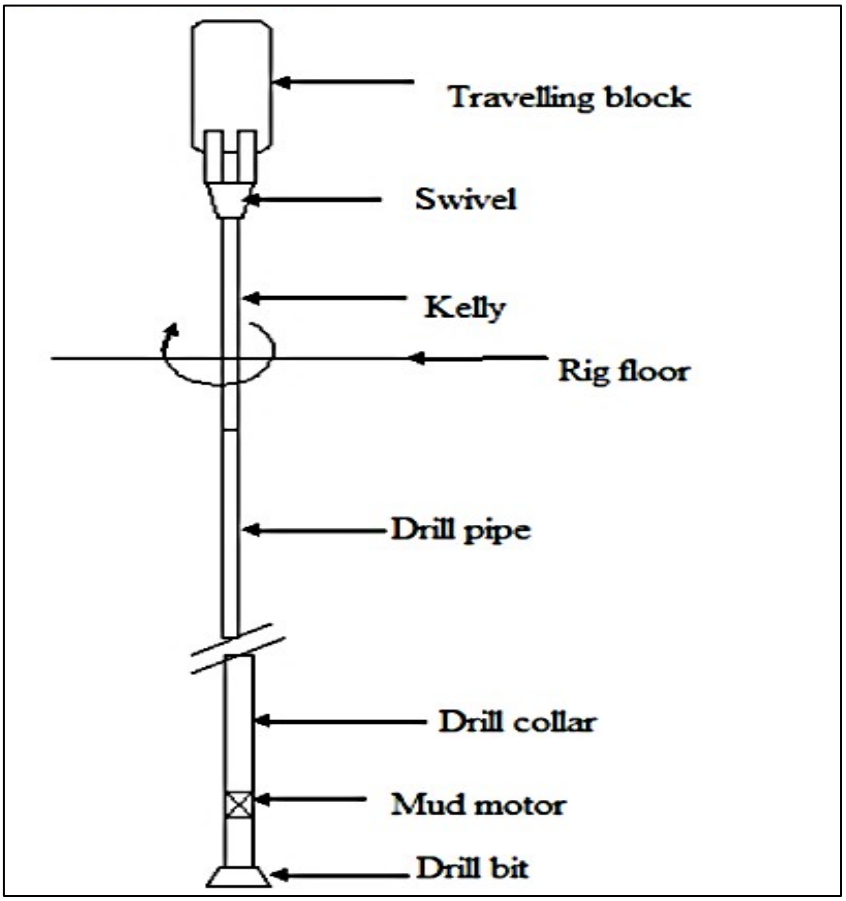

Figure Chapter Three .12 Convectional rig drill string

3.6 Mud Circulation System

The mud circulation system is the unsung hero of the drilling process, fulfilling several critical functions. This closed-loop system pumps drilling fluid, commonly referred to as mud, down the drill string, through the bit, and back up the wellbore [30].

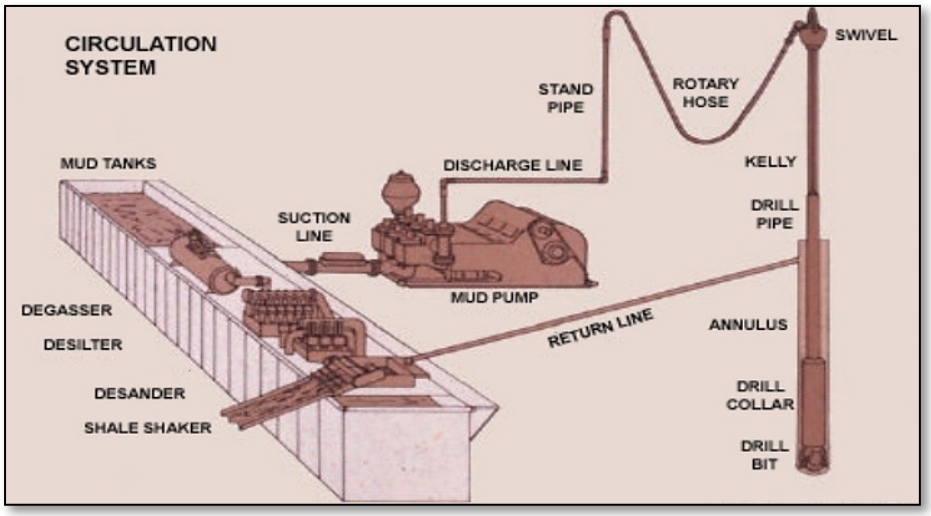

Figure Chapter Three .13 Mud Circulation System.

3.6.1 Mud Composition:

Drilling mud is a carefully formulated mixture of water, clays, weighting agents, and various additives. The specific composition varies depending on the geological formation and drilling conditions [17].

3.6.2 The mud performs a variety of essential tasks:

Drilling mud plays a crucial role in the success of any drilling operation, performing several essential tasks beyond just removing rock cuttings from the wellbore. Here's a summarized list of its key functions [28]:

- **Cool and lubricate the drill bit**: Reducing friction and heat buildup to prevent premature wear and tear.
- **Clean the wellbore**: Carrying rock cuttings away from the bit and preventing them from clogging the drilling process.

- **Maintain wellbore pressure**: Balancing formation pressures and preventing fluids from entering the wellbore, ensuring well control.
- **Stabilize the wellbore wall**: Depositing a thin mud cake on the borehole wall to prevent cave-ins.

3.6.3 Mud Circulation System Components:

The main components related to the circulating system are [23]:

- **Mud Pits**: Large tanks where the drilling mud is mixed and stored.
- **Mud Pumps**: Powerful pumps that push the mud through the entire system.
- **Standpipe and Rotary Hose**: Flexible connection delivering mud from the pumps to the kelly.
- **Swivel**: A rotating joint allowing the drill string to rotate while maintaining a continuous mud flow.
- **Drill String**: Provides a conduit for mud flow downhole and back up the wellbore.
- **Shaker Screens**: Vibrating screens that separate rock cuttings from the returning mud.
- **Mud Cleaners and Centrifuges**: Further refine the mud by removing finer particles and contaminants.

3.7 Well Control Equipment System

We've explored the key players in the drilling rigs orchestra - the power, hoisting, rotary, and mud circulation systems. Now, we meet the silent sentinels: the well control equipment system. This critical system operates behind the scenes, guarding against the ever-present threat of uncontrolled hydrocarbon flow, also known as a blowout [31].

3.7.1 The Significance of Well Control

Imagine that a volcano, miles beneath the surface of the earth, is about to erupt. In essence, that is what a blowout is. Personnel, the environment, and even the climate are at risk from the uncontrolled pressure coming from the wellbore, which can discharge oil, gas, and hazardous chemicals [30].

3.7.2 Layers of defence

The well control equipment system forms a multi-layered defence against blowouts, preventing, detecting, and controlling wellbore fluids. It comprises an array of specialized equipment, categorized into [13, 25]:

A. Primary Well Control Equipment

When it comes to primary well control equipment components, we can further break them down into two main categories based on their location and function:

1) **Blowout Preventer (BOP) Stack** The heart of the system, a series of powerful hydraulically-operated valves installed at the wellhead. Different types of BOPs cater to various functions, including:
   - Annular BOPs: Seal around the drill pipe or casing to prevent uncontrolled flow.
   - Ram BOPs: Close off the wellbore entirely using shearing or sealing rams.
   - Double BOPs: Redundant system with two BOPs of the same type for added safety.

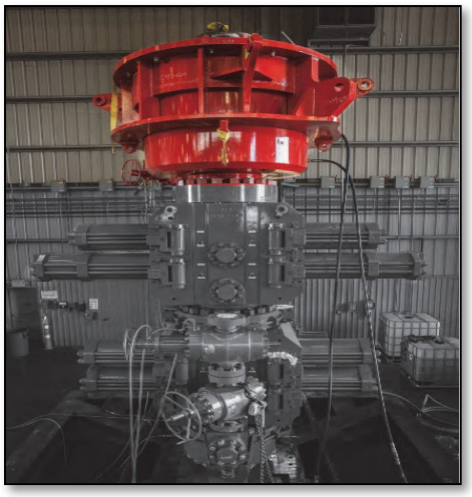

Figure Chapter Three .14 Annular preventer.

2) **Kelly Cock:** A valve located on the kelly that can quickly shut off drilling fluid flow in case of a kick (initial influx of formation fluids) [24].

B. Secondary Well Control Equipment:

- Diverter System: Channels uncontrolled flow away from the wellbore and rig through a separate pathway, minimizing environmental impact.
- Mud Gas Separator: Separates gas from the returning drilling mud, providing early warning signs of a kick [13].

C. Tertiary Well Control Equipment:

- Kill Lines and Pumps: Inject high-density drilling fluid down the wellbore to overcome formation pressure and regain control.
- Casing Cutters and Well Abandonment Equipment: Used in extreme situations to sever the wellbore and isolate it permanently [31].

4 Chapter Four

4.1 Material and Methodology

Drilling rig procedures can be quite complex and vary depending on the specific type of rig, well depth, formation being drilled, and desired outcome. However, there's a general outline of key steps involved in most drilling operations:

a. We began working on the project's design, making several sketches and graphs until we reached the final design.

b. The first time, we collected some iron pieces, selected a shape, and created.

Figure Chapter Four .15 Iron elements are assembled to achieve the predetermined design.

c. Then, to create our table, we took an old door and put four little legs and four wheels.

d. To improve the project's value, we decided to use car seat raisers instead of usual draw-work.

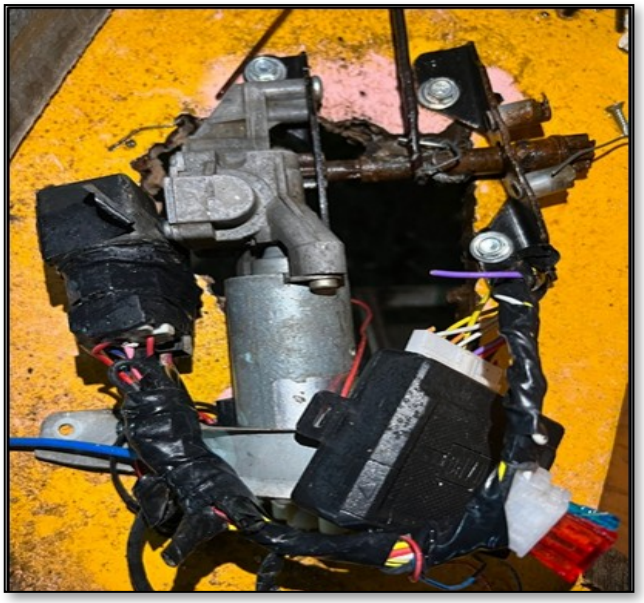

Figure Chapter Four .16 The utilization of car seat raisers as drawworks

e. In addition, we used a little wheel in place of the crown block.

Figure Chapter Four .17 A little wheel was used in place of the typical crown block.

f. We used a car radiator fan as a rotor to rotate the drill bit and our drill pipe.

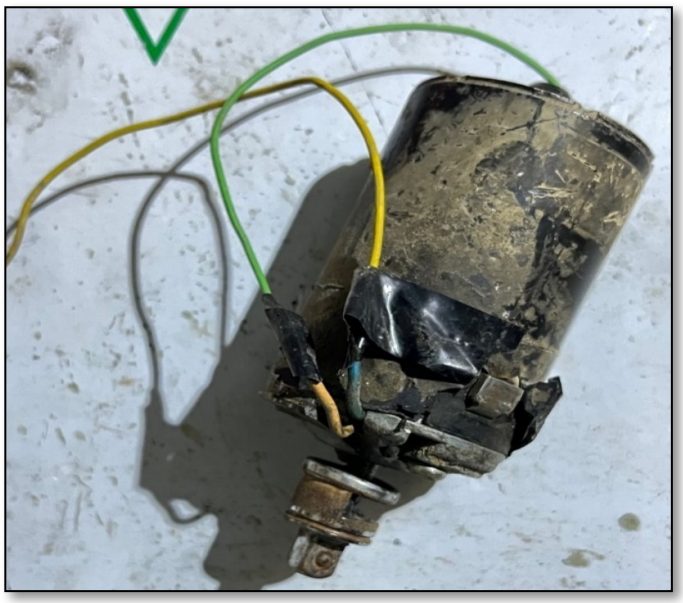

Figure Chapter Four .18 Our drill bit and drill pipe were rotated by means of a car radiator fan

g. Instead of drill pipes, we installed house water pipes.

Figure Chapter Four .19): House water pipes were erected in place of drill pipes.

h. The drill bit is replaced by the rings within the wheel truck.

Figure Chapter Four .20): The Utilization of the inner rings from a wheel truck as a drill bit.

i. An additional important component of the project was the installation of a vehicle wash tank as a mud tank. We also included a vehicle wash motor in our design to power the drilling fluid pump.

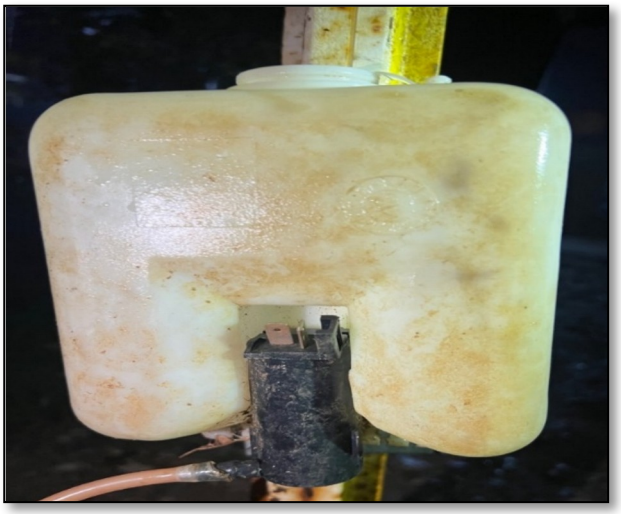

Figure Chapter Four .21 A vehicle wash tank converted into a mud tank.

j. Finally, as a source of energy for our project, we have also installed car batteries.

k. We started coloring our project after finishing these activities. We decided to have a black table and a blue derrick. Additionally, we designed the drill bit's gold color.

Figure Chapter Four .22 The initial stage of our rig's coloring.

5 Chapter Five

5.1 Conclusion

We know that we used two techniques in drilling which are new and old techniques. Also we have some systems in drilling technique that each one of them helps drilling and it is so important to drill a well safely and in an economical way. And also we can drill a well in the land and also in the sea that each one of them needs different techniques and different costs also it is possible to drill a well in vertical way.

5.2 Discussion (Future Work)

We are suggesting in future working more on drilling rig so more became detail.

Firedamp all component and all part circulation system and well controlling system, also to care by rotary drilling and discussing all difficulties during drilling will happened.

پـــوخـــتـــه

نەوت کە هەندێک جار بە نەوتی خاو یان تەنها نەوت ناودەبرێت، تێکەڵەیەکی شلەی سروشتییە و بە شێوەیەکی بنچینەی لە هایدرۆکاربۆنەکان پێکهاتووە کە لە پێکهاتە جیۆلۆجیەکاندا دۆزراوەتەوە. لەناو سەرچاوە وزەکاندا، نەوت بە تایبەتی گرنگە. کە

،تەکنیکی سەرەکی بۆ دەرهێنانی نەوت لە ژێر ڕووی زەوی، هەڵکەندن یاخود لێدانی بیری نەوتە

هەموو ئەو کارانە دەگرێتەوە کە لەسەر بیرەکە ئەنجام دەدرێ لە دەستپێکەوە هەتاکوو گەیاندنی .بە دەستی تیمی یاخود دەستەی بەرهەمهێنان

ئەم توێژینەوەیە ئامانجی هەڵسەنگاندنی وردە بۆکاریگەرییە ژینگەییەکان و ڕوونکردنەوەی ئەو .کار و پرۆسە سەرەکیانەیە کە لە کاتی هەڵکەندنی بیرە نەوتەکاندا ئەنجام دەدرێن

ئەمانە ڕێکارگەلێکی مەترسیدار و وەبەرهێنانی دارایین کە قازانج لە سەرووی هەموو شتێکەوە پەیوەستە بە کارایی و سەلامەتی. قەبارەدانان و بەڕێوەبردنی دروستی ئەم پرۆسانە زۆر گرنگە بۆ

سەرکەوتنی پڕۆسەی هەڵکەندنی بیرەکان. لەم ڕوانگەیەوە شرۆڤەکردنی ڕێکارەکان و گرنگی و بایەخی کارکردن کە پەیوەندییان بەم پرۆسانەوە هەیە، دەبنە شتێکی چارەنووسساز بۆ پیشەسازی .نەوت و گاز و بۆ کۆمەڵگا

بەو پێیەی لە ئێستادا نەوت وەک سەرچاوەی سەرەکی وزە بەکاردەهێندرێت ، یارمەتیدەرە لە .پەرەپێدانی باشترین پراکتیزەکان و پێشخستنی تەکنەلۆژیای هەڵکەندنی بیرەکان

لە توێژینەوەکەدا ئامرازە سەرەتاییەکان کە لە پرۆسەی هەڵکەندنی بیرێکدا بەکاردەهێنرێن سەرەتا باس دەکرێن بۆ زانیاری باشتر لەسەر هەموو کارەکان ، دواتر کارەکانی ماوەی هەڵکەندن بە شێوەیەکی گونجاو باس دەکرێن. پڕۆگرامی هەڵکەندن کە وەسفێکی وردی هەموو کارێک .لەخۆدەگرێت لە کۆتایی پرۆسەی تەواوکردنی بیرەکەدا پێشکەش دەکرێت

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